

IT-416 Computer Vision Project

Team 21 : Aadharsh Ramachandran(231IT001), Jaivant Kosaraju(231IT028), Paluvadi Dinesh Manideep(231IT047) & Sanga Balanarasimha(231IT062)

Project title : VMI-ATN-RCNN: A hybrid deep learning model for fish disease segmentation and classification in aquaculture

Contributions :

Jaivant Kosaraju – Data pipeline, training, and evaluation

Prepares the dataset by implementing preprocessing, augmentation, and the DataLoader. Also builds the training loop, loss functions, and evaluation metrics to train and test the final VMI-ATN-RCNN.

Paluvadi Dinesh Manideep – VMINet backbone

Implements the hybrid backbone network that combines VGG-16, MobileNet-V2, and Inception-V3 to extract multi-scale feature maps from fish images. These features are passed to the detection network.

Sanga Balanarasimha – Attention-based RPN and integration

Develops the Region Proposal Network with an attention-based multi-scale convolution layer to focus on disease-affected regions. Integrates this detection module with the VMINet backbone to generate bounding boxes for infected areas.

Aadharsh Ramachandran – Crayfish Optimization Algorithm (COA)

Implements the Self-Adaptive Crayfish Optimization Algorithm to automatically tune hyperparameters such as learning rate and batch size. This optimization improves the model's training efficiency and final accuracy.